INDUSTRIAL PHYSICS & ROBOTICS CURRICULUM

A First-Principles Academic Roadmap for Sovereign Builders

YEAR 1: FOUNDATIONAL MATHEMATICS & CLASSICAL MECHANICS

Semester 1

1. Calculus I: Single-Variable Differential & Integral Systems

Core Topics: Limits, derivatives, optimization dynamics, mean value theorem, fundamental theorem of calculus, integration techniques, numerical approximations, sequences, and infinite series.

Textbook: *Calculus: Early Transcendentals* by James Stewart

2. Classical Physics I: Mechanics & Particle Kinematics

Core Topics: Vectors, Newton's Laws of Motion, work-energy theorem, conservation of linear momentum, rotational kinematics, angular momentum, gravitation, and harmonic oscillations.

Textbook: *University Physics with Modern Physics* by Young and Freedman

Semester 2

3. Calculus II: Multivariable Vector Fields & Transformations

Core Topics: Parametric equations, polar coordinates, vector functions, partial derivatives, directional gradients, Lagrange multipliers, multiple integrals, Green's, Divergence, and Stokes' Theorems.

Textbook: *Calculus* by Tom M. Apostol

4. Classical Physics II: Electromagnetism & Field Theory

Core Topics: Coulomb's Law, Electric Fields, Gauss's Law, capacitance, resistance, DC/AC circuits, Magnetic Fields, Ampere's Law, Faraday's Law, and Maxwell's Equations.

Textbook: *Physics for Scientists and Engineers* by Serway and Jewett

YEAR 2: STRUCTURAL ANALYSIS, CIRCUITRY & COMPUTATIONAL LANGUAGE

Semester 3

5. Ordinary Differential Equations (ODEs) & Linear Systems

Core Topics: First-order equations, linear second-order equations, Laplace transforms, autonomous linear systems, phase portraits, stability analysis, and mathematical modeling.

Textbook: *Elementary Differential Equations* by Boyce and DiPrima

6. Analytical Mechanics & Wave Dynamics

Core Topics: Lagrangian and Hamiltonian mechanics, generalized coordinates, principle of least action, central force problems, coupled oscillations, fluid statics, and mechanical wave equations.

Textbook: *Classical Mechanics* by Herbert Goldstein

Semester 4

7. Linear Algebra & Vector Space Matrix Optimization

Core Topics: Systems of linear equations, matrix algebra, determinants, vector spaces, basis, dimension, inner products, linear transformations, eigenvalues, eigenvectors, and SVD.

Textbook: *Introduction to Linear Algebra* by Gilbert Strang

8. Industrial Electrodynamics & Electronic Systems

Core Topics: Semiconductor basics, diodes, BJTs, FETs, operational amplifiers, analog filters, digital logic gates, combinatorial logic, circuit transient analysis, and hardware design layout.

Textbook: *The Art of Electronics* by Horowitz and Hill

YEAR 3: THE ARCHETYPE FRAMEWORK — THERMAL, FLUID & QUANTUM SYSTEMS

Semester 5

9. Thermodynamics & Statistical Mechanics

Core Topics: Laws of thermodynamics, heat engines, entropy, thermodynamic potentials (Gibbs/Helmholtz), chemical equilibrium, phase transitions, and Maxwell-Boltzmann distributions.

Textbook: *Thermal Physics* by Charles Kittel and Herbert Kroemer

10. Fluid Dynamics & Transport Phenomena

Core Topics: Navier-Stokes equations, inviscid flow, continuity equations, Bernoulli's equation, laminar/turbulent flows, Reynolds number, boundary layer theory, and heat/mass transport formulas.

Textbook: *Introduction to Fluid Mechanics* by Fox, McDonald, and Pritchard

Semester 6

11. Foundational Quantum Mechanics I

Core Topics: Wave-particle duality, Bohr model, wave functions, Born interpretation, Uncertainty Principle, Time-independent Schrodinger equation, harmonic oscillators, and operators.

Textbook: *Introduction to Quantum Mechanics* by David J. Griffiths

12. Optics, Laser Physics & Industrial Photonics

Core Topics: Geometrical optics, wave optics, interference, diffraction, polarization, laser principles, fiber optics, optoelectronic modulators, photodetectors, and industrial laser processing.

Textbook: *Optics* by Eugene Hecht

YEAR 4: INDUSTRIAL SYNTHESIS, SEMICONDUCTORS & ADVANCED ROBOTICS

Semester 7

13. Solid-State & Semiconductor Microchip Physics

Core Topics: Crystal lattices, reciprocal lattice vectors, Bloch's theorem, energy bands, metals/semiconductors, intrinsic/extrinsic densities, p-n junctions, and transistor mechanics.

Textbook: *Introduction to Solid State Physics* by Charles Kittel

14. Robotics I: Kinematics, Dynamics & Spatial Geometry

Core Topics: Spatial transformations, Denavit-Hartenberg parametrization, forward/inverse kinematics of serial link manipulators, Jacobian matrices, singularity identification, and trajectory generation.

Textbook: *Introduction to Robotics: Mechanics and Control* by John J. Craig

Semester 8

15. Robotics II: Feedback Control Systems & State Estimation

Core Topics: Open/closed loop dynamics, PID tuning, state-space representations, controllability, observability, Lyapunov stability, Kalman filters, sensor fusion, and autonomous trajectory planning.

Textbook: *Feedback Control of Dynamic Systems* by Franklin, Powell, and Emami-Naeini

16. Advanced Materials Science & Automated System Design

Core Topics: Mechanical properties (stress-strain, fatigue, creep), polymers, ceramics, composite matrix design, actuators, embedded microcontrollers, and automation system integration.

Textbook: *Materials Science and Engineering* by William D. Callister